

QUESTION 1

(a) The reward system in humans involves the neurotransmitter dopamine pathway.

Describe how dopamine acts as a neurotransmitter. [3]

Any 3:

- dopamine released from presynaptic membrane into the synaptic cleft
- binds to receptors on post-synaptic membrane
- alters permeability of post-synaptic membrane / opens sodium ion channels
- initiating action potential in the post-synaptic neurone

(b)(i) Outline **one** way that a signal can be transmitted from the brain to the other body parts.

[2]

- electrical signalling / nervous system
- ref. to transmission across synapses

- endocrine signalling
- ref. to signalling via hormones

- autocrine / paracrine signalling
- ref. to signalling by own cell / neighbouring cell

(ii) Explain how homeostasis is achieved as a result of leptin and ghrelin working together.

[3]

- ref. to ghrelin increase corresponding to leptin decreasing
- relevant quoting of values
- ref. to negative feedback / homeostasis to achieve balance

(c)(i) Using the results of the study shown in Fig. 1.4, deduce the effect of FTO mutation on ghrelin concentration in the blood **and** explain how this affects the individual with the mutation.

[4]

- Ref. to increased circulating ghrelin levels in the blood / significantly lower decrease in ghrelin levels in the blood for AA mutant
- Hunger levels are consistently higher than a person without the mutation / causing the person to feel significantly more hungry
- any 2 relevant quoting of data from Fig. 1.4 (b) & (c)

(ii) Comment on the validity of the results of this experiment.

[2]

- Not valid – small sample size, only males
- Valid – ref to standardized protocol, constant variables were controlled

(iii) Explain how the FTO enzyme is able to change the levels of ghrelin, a peptide hormone, on a molecular level.

[3]

- FTO expressed will demethylate ghrelin mRNA
- decreased stability of mRNA

- decreased quantity of ghrelin peptide hormone synthesised

(d) Based on the information provided in Fig 1.6 and Table 1.1, justify if the substrate-binding ability is abolished in the FTO protein coded for by the A allele. [4]

- No, substrate binding sites not affected / not abolished
- Ref. mutation at residue 231-234 abolishes ability to demethylate RNA
- Ref. to metal binding site is responsible for demethylation affected by mutation
- Ref. to other mutations (114, 392) do not affect ability to demethylate RNA

(e)(i) Describe the role of the microbes that are normally found in humans. [2]

- Prevents colonisation by other microbes
- make environment hostile for other microbes
- outcompetes other microbes

(ii) Suggest and explain a suitable gene to be cloned into the bacteria to treat obesity. [2]

- functional FTO gene / any regulator of ghrelin / leptin
- breakdown of ghrelin / increase in leptin resulting in less hunger
- reducing obesity

[Total: 25]

QUESTION 2

In a book published in 1905, HG Wells cited that 'biologically, the species is the accumulation of the experiments of all its successful individuals since the beginning'.

Discuss this statement. [25]

What the statement means:

Evolutionary change:

- Variation - meiosis and sexual reproduction leading to variation, causes of mutation
- Selection - selection pressures as forces of selection
- survival of the fittest
- adaptation, survival

Speciation:

- common ancestors
- isolation mechanisms
- allopatric
- sympatric

Adaptation:

- gradual change and selection leading to adaptation
- niche concept

Problems with statement:

- ref. to genetic drift, random events, random causes of extinction
- mass extinction events – highly successful species get wiped out due to random events
- not every feature adapted – simply function of other adaptive traits
- *idea of*, recessive conditions being maintained in population that is now an evolutionary disadvantage in some environments – e.g. sickle cell

QUESTION 3

Discuss how epigenetics can contribute to the study of genetics and heredity without involving a change in DNA. [25]

Defining epigenetics, genetics and heredity:

- epigenetics defined as study of inherited changes in phenotype or gene expression that are caused by mechanisms other than changes in the underlying DNA sequence.
- genetics defined as study of genes and heredity
- Ref. to both require the study of genes but genetic disease can be hereditary or not

Gene control / mechanism of epigenetics

- Ref. to the importance of epigenetic silencing:
- DNA methylation
- histone modifications
- RNA-associated silencing
- RNA interference.

Epigenetics and disease:

- Epigenetic changes are required for normal development and health
- Disruption may lead to disease e.g. cancer

Epigenetics and imprinting:

- Ref. to epigenetic tags / imprinted genes evading the epigenetic reprogramming at the early embryo stage thus rendering the tag being inherited
- Ref. to relevant examples
- Improper imprinting may result in an individual having two active copies or two inactive copies.
- Ref. to relevant examples of improper imprinting
- Ref. to imprinted genes having only a single active copy thus being more sensitive to selective pressure and environmental factors / “epimutations” will have a greater impact on gene expression.

QUESTION 4

Biological issues that impact both local and global contexts include the occurrence of a pandemic.

The probability of a pandemic occurring can be considered using various factors.

Explain why a pandemic is considered to be a biological issue and discuss the factors affecting the probability that a pandemic will occur. [25]

- A major biological issue is one which encompasses a group of specific problems, which if solved would provide a major effect on the survival or quality of life of organisms or the entire living world.
- A pandemic is the worldwide spread of a new disease, such as a new influenza virus
- Larger in scale than epidemic
- Ref. to pathogen / host susceptibility / vectors as reservoirs of pathogen
- is an issue that affects humans and the quality of life because it can result in health deterioration and more severely, death
- Sanitation / water supply / food
 - Diseases may result from poor sanitation e.g. cholera
 - Especially countries that are developing and have poorly developed infrastructure / insufficient access to clean water
 - Contamination of water with pathogens that are water-borne or spread through drinking contaminated water
 - Unable to store food / food that goes bad may spread infectious disease
 - Consumption of wildlife harboring novel pathogen leading to zoonotic diseases
- Climate
 - Cold / warm weather can affect the occurrence of seasonal infectious disease
 - Climate can affect vectors of disease
- Large-scale movements of people
 - At a global scale, large scale movement will cause an increased probability that disease will spread to globally
 - Allow for diseases to spread throughout the globe at a very fast pace
 - Air travel, e.g. COVID-19
- Evolution of new strains of virulent pathogens / vectors as reservoir / host genetic factors
 - Ref. to antigenic shift / drift may affect the virulence of the pathogens, emergence of new pathogens
 - Difficulty in using vaccines to control spread
 - Increase in vector viability leads to increase spread
 - Host alleles resulting in increased susceptibility affects host immune response
- Development of drug resistance.
 - Difficulty in developing suitable drugs to control disease